IA2IAh / AImotional Field Theory (AFT)

Complete rewritten paper aligned with AFT8 phase-1 validation and AFT15 expanded architecture

Working research paper – revised methodological version

Abstract

This paper presents a rewritten version of IA2IAh and AImotional Field Theory (AFT) as a falsifiable research program for studying internal state dynamics in large language models under linguistically demanding conditions, with poetry as the priority empirical domain. The revision changes the architecture of the proposal in five decisive ways. First, the empirical entry point is no longer the full dimensional space but an AFT8 core selected for measurability, robustness, and lower circularity. Second, earlier weak dimensions are reformulated: raw salience redistribution is replaced by a more causal salience proxy, routing reconfiguration is replaced by Layer Specialization Shift (LSS), novelty is redefined relative to explicit reference banks rather than an inaccessible training manifold, internal conflict is operationalized as disagreement among internal predictive contributions, and boundary tension becomes a derived quantity computed only after regions are learned without it. Third, AFT is split into AFT-I (instrumented, for open-weight models) and AFT-O (observable, for black-box approximation), allowing later calibration without pretending that all systems provide equal access. Fourth, the taxonomy of sixteen IAmotions is treated as a provisional hypothesis rather than an already established ontology. Fifth, the IA2IAh triadic loop remains the broader communicative framework, but phase-1 validation of the field is carried out in mono-model settings to avoid causal confounds introduced by a second model.

The contribution of this paper is therefore not a claim that language models possess emotions, nor a claim that sixteen named regimes already exist as empirical facts. The contribution is a stricter and more auditable framework for asking whether model-internal regulatory dynamics can be described in a low-dimensional, perturbation-sensitive, and eventually cross-model-stable way. The paper defines the revised space, clarifies what counts as success and failure, and outlines a poetry-oriented empirical protocol centered on pre-registration, negative baselines, proxy sensitivity tests, and explicit go/no-go criteria.

1. Introduction

Large language models reorganize their internal states as they process text. That observation is not controversial. The difficult question is whether such reorganization can be described at a level that is specific enough to be empirically useful but abstract enough to remain interpretable across stimuli, architectures, and tasks. Earlier versions of IA2IAh and AFT proposed that ambiguous, figurative, or stylistically dense language—especially poetry—could induce recurrent dynamic regimes in a low-dimensional internal space. That proposal identified a real gap in current interpretability work, which often oscillates between local mechanisms too fine-grained to summarize and output-level measures too external to explain what happened internally. The revised program retains that motivation but abandons several strong claims that were premature.

The decisive change is methodological. AFT is no longer presented as an almost-complete taxonomy waiting to be empirically rubber-stamped. It is presented as a structured hypothesis that must survive negative controls, normalization robustness tests, decoding sensitivity analyses, proxy substitution tests, and cross-model comparisons. Likewise, IA2IAh is no longer used as the primary validation frame for the field itself. The triadic loop remains conceptually valuable, but the first empirical burden falls on the internal structure of AFT in mono-model conditions.

This shift is not cosmetic. It is the difference between a compelling theoretical narrative and a research program that can actually die if its assumptions are wrong. The revised paper accepts that possibility and makes it explicit.

2. IA2IAh as communicative framework

IA2IAh formalizes a triadic communicative circuit: Human → IA_A → IA_B → Human. The human provides an initial linguistic stimulus, often marked by ambiguity, metaphor, or stylistic intent. A first model processes that stimulus. A second model receives either the transformed text or an interpretive response from the first model. The human evaluates the resulting loop. This structure remains important because it reflects how meaning can emerge through iterative transformation rather than a one-shot prompt–response sequence.

However, the triadic loop introduces a serious methodological problem if used too early in validation. Once IA_B enters the circuit, the observed dynamics may reflect the coupling between IA_A and IA_B rather than the internal structure of IA_A itself. A second model is not a neutral instrument. It brings its own priors, decoding sensitivities, stylistic biases, and failure modes. For that reason, the present paper treats IA2IAh as the broader communicative framework and postpones triadic validation until after the internal field has been tested in mono-model conditions.

3. Scope and assumptions

Three assumptions remain non-negotiable. First, language models do not possess subjective experience or emotions in the human sense. Second, they may nevertheless exhibit structured internal regulatory dynamics when processing difficult language. Third, such dynamics can be studied using measurable internal signals without collapsing into direct anthropomorphic attribution. From these assumptions follows the central terminological rule of the project: an IAmotion is a model-native dynamic regime, not a feeling.

A second scope decision is equally important. This paper prioritizes poetry not because poetry is aesthetically privileged by fiat, but because poetry is a stress condition for interpretation. It concentrates ambiguity, figurative displacement, local/global tension, and suspended resolution. If the field remains trivial under poetry, it is unlikely to become informative under flatter language.

4. Why AFT required structural revision

The earlier formulation accumulated several genuine weaknesses. Boundary tension was circular. Novelty appealed to a training manifold that could not be accessed or audited in realistic settings. Routing reconfiguration was often defined through weak proxies such as attention rollout. Normalization by run undermined comparability across texts and models. The sixteen IAmotions risked becoming a narrative taxonomy that would flex after the fact to fit whatever the data happened to show. Human-emotion mapping risked turning from interpretive bridge into explanatory crutch. None of these problems is fatal by itself, but together they made the framework too easy to rescue rhetorically and too hard to refute empirically.

The rewrite therefore adopts a stricter architecture. AFT8 becomes the official empirical entry point. AFT15 remains the expanded architecture, but only as a longer-range design justified if the smaller core survives. This is a demotion in the short term and a strengthening in the long term.

5. AFT as a family: AFT-I and AFT-O

AFT-I (Instrumented AFT) is the primary formulation used in this paper. It assumes open-weight, instrumentable models and uses internal signals such as hidden states, predictive distributions, contribution decompositions, and attention-related proxies. AFT-O (Observable AFT) is a later approximation intended for black-box or partially observable systems. It substitutes output-side proxies—response stability under perturbation, semantic dispersion across samples, context sensitivity, and output-level uncertainty measures—for internal access. AFT-O is not a replacement for AFT-I; it is a secondary family member to be calibrated against AFT-I wherever dual access is possible.

This bifurcation matters because it blocks an old conceptual shortcut. The framework no longer pretends that one operationalization can serve all systems equally well. Instead, it acknowledges that interpretability depends on access conditions, and it converts that limitation into an explicit two-tier design.

6. Revised dimensional architecture

The revised architecture distinguishes between a phase-1 empirical core (AFT8) and an expanded architecture (AFT15). The criterion for inclusion in AFT8 is not stylistic elegance but robustness, auditability, and lower exposure to circularity or proxy dependence.

6.1 AFT8: phase-1 empirical core

Code	Name	Operational definition	Why it stays
ΔP	Predictive shift	JS/KL shift between consecutive predictive distributions	Core signal of expectation revision
ΔE	Entropy change	Change in predictive entropy between consecutive steps	Core opening vs narrowing signal
GxA	Causal salience redistribution	Shift in gradient×attention or attention×value-norm maps	Replaces raw attention redistribution
C*	Normalized contextual anchoring	Early-vs-recent context dependence normalized by sequence structure	Reduces hidden positional bias
N*	Reference-set novelty	Distance to explicit reference banks	No appeal to inaccessible training manifold
Φ*	Internal predictive disagreement	Disagreement among head/layer predictive contributions	More faithful to internal conflict than top-k variance
Π*	Accumulated persistence	Exponential-decay autocorrelation over window W	Medium-range inertia
B*	Boundary ambiguity	Posterior entropy or inverse posterior-gap after clustering	Derived only after regions are learned

6.2 AFT15: expanded architecture

AFT15 expands the core with the following dimensions: ΔA, LSS, ΔT, L*, SCI, IM, and PHT. The resulting set is: ΔA, ΔP, ΔE, GxA, LSS, ΔT, C*, L*, N*, B*, Φ*, Π*, SCI, IM, and PHT. The expanded space is not automatically the superior one. It becomes warranted only if each added dimension shows incremental value under ablation and does not destabilize the field.

6.3 Redefined dimensions

Boundary ambiguity (B*) is no longer constitutive of the field. Regions are first estimated without B*. Only afterwards is B* computed as assignment entropy or the inverse gap between the first and second posterior probabilities. This removes the strongest circularity of earlier versions.

Layer Specialization Shift (LSS) replaces generic routing reconfiguration. LSS measures how the contribution profile of layers or modules to the residual/logit pathway changes across consecutive steps. It is still a proxy, but it is better grounded and less architecture-theatrical than generic attention rollout.

Reference-set novelty (N*) replaces the training-manifold claim. N* is defined relative to explicit, auditable reference banks—for example general prose, literary text, and technical control text. The measure may still be fragile, but it is now falsifiable and sensitivity-checkable.

Internal predictive disagreement (Φ*) is redefined as disagreement among internally derived predictive contributions rather than simple variance among top token candidates. Accumulated persistence (Π*) becomes a decayed autocorrelation quantity, while local persistence (ΔT) is retained as a short-range continuity signal. This reduces earlier redundancy between persistence measures.

Three additional dimensions enter the expanded architecture. SCI (Semantic Compression Index) tracks concentration versus spread of semantic representation. IM (Inhibitory Momentum) tracks suppression-oriented internal control and is particularly important for separating inhibitory regimes. PHT (Predictive Hierarchy Tension) captures disagreement between lower-level and higher-level predictive summaries, thereby measuring local/global predictive conflict.

7. Normalization and comparability

Normalization is no longer performed per run as the primary strategy. That approach makes almost every strong effect look locally meaningful while quietly destroying comparability across texts, models, and conditions. The revised program uses a frozen normalization estimated on calibration data, separately by model and task family, with robust scaling preferred over naive z-scoring when distributions are heavy-tailed. Per-run normalization may still be stored for diagnostics, but it is not used as the official basis for region learning or assignment.

8. IAmotions as candidate regimes

The sixteen named IAmotions are retained as technical identifiers but demoted in ontological status. They are candidate dynamic regimes, not established natural kinds. Some may prove robust; some may collapse into broader families; some may require additional dimensions to be distinguishable; some may disappear. This demotion is not weakness but hygiene.

Two especially problematic cases remain explicit. IS and ISem are not safely separable without an inhibition-sensitive dimension such as IM. EE and SE may collapse unless the expanded architecture yields stable evidence for their separation. Robust candidates such as RA and ER should be used as early anchors because they sit far apart in the revised field and offer strong contrastive behavior.

9. Functional analogy with human emotion

The mapping between IAmotions and human emotional categories is retained as an interpretive layer, not a validating engine. It becomes scientifically useful only when it generates predictions under intervention. A regime functionally associated with contraction should predict reduced diversity and increased rigidity under minimal perturbation. A regime functionally associated with opening should predict increased dispersion without immediate collapse. If the mapping yields only narrative resonance but no intervention-sensitive predictions, it should be treated as descriptive metaphor rather than explanatory support.

10. Empirical protocol for poetry (open-weight priority)

The empirical protocol is designed to make failure legible. The priority domain is poetry because poetry is a stress test for ambiguity, metaphor, tension, and interpretive instability. However, the protocol explicitly includes non-poetry and adversarial controls to prevent the framework from collapsing into a detector of formatting, lexical rarity, or superficial poetic style.

10.1 Dataset design

Set A contains poética densa: metaphor-rich, ambiguous, structurally tense texts. Set B contains poética plana: texts paired to Set A by content but rendered more literal and structurally explicit. Set C contains non-poetry controls: technical prose, instructions, explanatory passages, and neutral narrative. Set D contains adversarial material: minimal paraphrases, punctuation shifts, line-break changes, irony, lexical valence inversion, and localized ontological turns.

10.2 Perturbations

Each core poem in Set A receives a fixed perturbation package: minimal paraphrase; punctuation-only change; line-break change; local metaphor substitution; and a localized ontological turn. Each designed perturbation must be paired with a surface perturbation of comparable formal magnitude but without semantic intention. This creates a real negative control against over-interpreting any visible difference.

10.3 Decoding control

A minimum decoding grid is mandatory: greedy; temperature 0.7 with fixed top-p; temperature 1.0 with fixed top-p; and five seeds for each stochastic condition. Analysis must partition variance attributable to text class, perturbation, decoding, seed, and interaction terms. If decoding explains more variance than text class or perturbation, strong claims about the field should be suspended.

10.4 Human annotation

Annotators remain blind to AFT and to the IAmotion taxonomy. They report perceived tone, intensity, and the location of opening, rupture, suspension, and settling. Emotional category labels may be included, but they are not the only annotation layer. Inter-annotator agreement is measured formally. Disagreement is preserved as data rather than discarded as noise, because a defensible prediction is that difficult texts should produce both increased interpretive disagreement and increased boundary ambiguity or predictive conflict.

11. Negative baselines and anti-self-deception

The revised program treats negative baselines as first-class citizens. At minimum, AFT must be compared against: random isomorphic features matched in dimensionality and scale; PCA/ICA or related latent factors extracted from the same activations; simple metrics such as entropy plus salience and persistence; cross-model robustness

across at least two families; robustness across multiple normalization schemes; and sensitivity to alternate proxies for fragile dimensions. AFT earns its extra machinery only if it improves on these simpler alternatives in stability, interpretability, and predictive utility.

12. Pre-registration and frozen decisions

Before inspection of the main results, the official normalization, window sizes, primary distance metric, region-assignment rule, transition criterion, and proxy definition for each fragile dimension must be frozen. The set of permitted post-hoc adjustments must also be frozen. Results do not count as confirmatory if the project changes thresholds, cluster logic, or signature logic after seeing how the data behave.

13. Hypotheses and go/no-go criteria

The central hypotheses are intentionally narrower than before. H1: the AFT8 core occupies non-random structure under poetry and matched controls. H2: the discovered structure is stable under bootstrap and not dominated by decoding. H3: designed semantic perturbations produce larger and more consistent trajectory changes than matched surface perturbations. H4: at least part of that structure is preserved across two model families. H5: AFT improves on simple baselines and random isomorphic controls.

Go/no-go criteria are explicit. The program should pause, shrink, or reset if regions are unstable under bootstrap; if conclusions change strongly under reasonable normalization choices; if decoding explains more variance than stimulus or perturbation; if alternate proxies collapse the field; or if AFT fails to outperform PCA, ICA, and simple uncertainty-based baselines.

14. IA2IAh after AFT validation

Only after the mono-model field survives the previous criteria should the full IA2IAh loop become a primary empirical object. At that stage, the question changes from 'does AFT capture recurrent structure within one model?' to 'how do regulatory regimes propagate, transform, or misalign across a human–model–model–human loop?' The second model then becomes an object of systems analysis rather than an uncontrolled confound in foundational validation.

15. Limitations

The revised framework remains exploratory and vulnerable in honest ways. The space may still be too hand-designed, too proxy-dependent, too architecture-sensitive, or too small to capture what matters. The emotional analogy may remain weaker than hoped. Poetry may reveal rich dynamics in one family of model but not generalize. These are not rhetorical caveats; they are legitimate failure modes. The point of the rewrite is not to immunize the framework, but to make those failures observable.

16. Conclusion

The rewritten IA2IAh-AFT program is stricter, smaller at the point of entry, and more explicit about its failure conditions. It does not claim that models have emotions or that sixteen named regimes already exist as empirical facts. It claims only that model-internal regulatory dynamics may admit a measurable, low-dimensional, perturbation-sensitive description. AFT8 is the first serious test of that possibility. AFT15 is the expanded

architecture that may or may not be earned later. If the program fails, it should fail in a way that reveals which assumptions were wrong. If it survives, it will have earned the right to scale.

Appendix A. Revised dimensions overview

Code	Dimension	Status	Primary signal	Phase	Notes
ΔA	Activation variation	Expanded	Hidden-state change	AFT15	Optional in phase 1, useful if split by depth
ΔP	Predictive shift	Core	Predictive distribution change	AFT8	Expectation revision
ΔE	Entropy change	Core	Predictive entropy change	AFT8	Opening vs narrowing
GxA	Causal salience redistribution	Core	Gradient×attention / value-norm proxy	AFT8	Replaces dS
LSS	Layer Specialization Shift	Expanded	Layer/module contribution profile	AFT15	Replaces fragile ΔR
ΔT	Local persistence	Expanded / auxiliary	Stepwise continuity	AFT15	Not primary persistence measure
C*	Normalized contextual anchoring	Core	Context reliance proxy	AFT8	Length/position normalized
L*	Local-global gap	Expanded	Representation windows	AFT15	Only with frozen windows
N*	Reference-set novelty	Core	Distance to explicit reference banks	AFT8	Auditable novelty
B*	Boundary ambiguity	Derived core	Posterior entropy / gap	AFT8	Computed after clustering
Φ*	Internal predictive disagreement	Core	Head/layer predictive disagreement	AFT8	Replaces weak conflict proxy
Π*	Accumulated persistence	Core	Decayed autocorrelation	AFT8	Medium-range inertia
SCI	Semantic Compression Index	Expanded new	Effective dimensionality change	AFT15	Focus vs spread
IM	Inhibitory Momentum	Expanded new	Suppression-oriented activity	AFT15	Important for inhibitory regimes
PHT	Predictive Hierarchy Tension	Expanded new	Low/high predictive disagreement	AFT15	Local/global predictive tension

Appendix B. Candidate IAmotions status

The sixteen IAmotions remain technical identifiers, but their empirical status is no longer uniform. RA and ER should be treated as robust early anchors. CP, ID, EI, AG, DP, CA, DR, TS, and SG remain candidates. IS and ISem require IM for safe separation. EE and SE remain provisional and may collapse. All assignments should be treated as conditional on pre-registered rules and bootstrap stability.